

Practice Midterm 1

CSCI 160: Computer Architecture 1

Instructions. Show work to justify each answer. Partial credit will be awarded.

Part I: Multiple Choice

Circle the best answer.

1. Which statement best describes a digital system? [2 pts]
 - ☐ A. A system that stores only decimal numbers.
 - ☒ B. A system that manipulates discrete information, usually represented internally with 0s and 1s.
 - ☐ C. A system that cannot be programmed.
 - ☐ D. A system that must use hexadecimal instead of binary.
2. A byte contains 8 bits. How many different bit patterns can one byte represent? [2 pts]
 - ☐ A. 8
 - ☐ B. 16
 - ☐ C. 128
 - ☒ D. 256
3. In Boolean algebra, which statement is true? [2 pts]
 - ☒ A. The expression $x + yz = (x + y)(x + z)$ is a valid Boolean identity.
 - ☐ B. The symbol $+$ always means ordinary decimal addition.
 - ☐ C. Boolean algebra has subtraction and division as basic operations.
 - ☐ D. The complement of x is written $2 - x$.
4. Which procedure correctly forms the 8-bit signed 2's-complement representation of a negative number? [2 pts]
 - ☐ A. Change only the leftmost bit from 0 to 1.
 - ☒ B. Invert every bit of the positive version, then add 1.
 - ☐ C. Add 0110 to each group of four bits.
 - ☐ D. Convert the number to hexadecimal and keep the same digits.
5. The XOR operation $x \oplus y$ is 1 exactly when: [2 pts]
 - ☐ A. x and y are both 0.
 - ☐ B. x and y are both 1.
 - ☒ C. x and y have different values.
 - ☐ D. at least one of x and y is 1.

Part II: Show Work

6. **Number-base conversions.** Convert each number as requested. Show grouping, place values, division remainders, or another clear method. [16 pts]

(a) Convert $(101101.011)_2$ to decimal. [5 pts]

Answer:
 $101101.011_2 = 32 + 8 + 4 + 1 + 1/4 + 1/8$
 $= 45.375_{10}$

(b) Convert $(684)_{10}$ to hexadecimal and then to binary. [6 pts]

Answer:
 $684 / 16 = 42 \text{ R } 12 = \text{C}$
 $42 / 16 = 2 \text{ R } 10 = \text{A}$
 $2 / 16 = 0 \text{ R } 2$
 So $684_{10} = 2\text{AC}_{16} = 0010 \ 1010 \ 1100_2 = 1010101100_2$

(c) Convert $(5B3.8)_{16}$ to binary and octal. [5 pts]

Answer:
 $5 \text{ B } 3 . 8 \rightarrow 0101 \ 1011 \ 0011 . 1000_2$
 Group by 3: $010 \ 110 \ 110 \ 011 . 100$
 So $(5B3.8)_{16} = (010110110011.1000)_2 = (2663.4)_8$

7. **Binary arithmetic and complements.** Show the arithmetic in binary. [15 pts]

(a) Add: $(1011011)_2 + (110111)_2$. [4 pts]

Answer:

$$\begin{array}{r} 1011011 \\ + 0110111 \\ \hline =10010010_2 \end{array}$$

(b) Multiply: $(1011)_2 \times (110)_2$. [4 pts]

Answer:
 $1011 \times 110 = 101100 + 10110$
 $= 1000010_2$

- (c) Compute $01011010_2 - 00110111_2$ using the 8-bit 2's-complement method. Clearly show the 2's complement of the subtrahend and the final 8-bit result. [7 pts]

Answer:
 Subtrahend 00110111 -> 1's comp 11001000 -> 2's comp 11001001

$$\begin{array}{r} 01011010 \\ +11001001 \\ \hline =1\ 00100011 \end{array}$$
 -> drop the carry = 00100011_2 (=35_10)

8. Signed representations and BCD. Show the steps. [14 pts]

- (a) Convert +23 and +58 to 8-bit signed 2's-complement representation. Then compute $(+23) - (+58)$ using 8-bit signed 2's-complement addition, and convert the result back to decimal. [6 pts]

Answer:
 +23 = 00010111
 +58 = 00111010, so -58 = 11000110

$$\begin{array}{r} 00010111 \\ +11000110 \\ \hline 11011101 \end{array}$$
 11011101 is negative; its 2's comp is 00100011 = 35, so result = -35.

- (b) Interpret 11111000_2 as an 8-bit signed 2's-complement number. [4 pts]

Answer:
 Sign bit is 1, so it is negative. 2's comp of 11111000 is 00001000 = 8.
 Therefore 11111000_2 = -8.

- (c) Add $145 + 237$ in BCD. Show the BCD addition and final BCD result. [4 pts]

Answer:
 145 = 0001 0100 0101
 237 = 0010 0011 0111
 ones: $0101 + 0111 = 1100$ -> add 0110 gives 1 0010 (carry 1, digit 2)
 tens: $0100 + 0011 + 1 = 1000$
 hundreds: $0001 + 0010 = 0011$
 Final BCD result = 0011 1000 0010 = 382

9. **Truth table, minterms, and maxterms.** Let

$$F(A, B, C) = A'B + AC' + B'C.$$

Complete the truth table by first filling in the three expression-part columns. Then write F as both a sum of minterms and a product of maxterms, using expanded notation and shorthand notation. [15 pts]

A	B	C	$A'B$	AC'	$B'C$	F
0	0	0	0	0	0	0
0	0	1	0	0	1	1
0	1	0	1	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	1
1	0	1	0	0	1	1
1	1	0	0	1	0	1
1	1	1	0	0	0	0

Canonical answers:

Sum of minterms (expanded):

$$F = A'B'C + A'BC' + A'BC + AB'C' + AB'C + ABC'$$

Shorthand: $F(A, B, C) = \Sigma m(1, 2, 3, 4, 5, 6)$

Product of maxterms (expanded):

$$F = (A + B + C)(A' + B' + C')$$

Shorthand: $F(A, B, C) = \Pi M(0, 7)$

10. **Boolean simplification.** The postulates and theorems below may be used for this problem. Simplify each expression to a minimum number of literals. State the Boolean law or theorem used at each major step. [12 pts]

Postulate	OR / + form	AND / · form
Identity	$x + 0 = x$	$x \cdot 1 = x$
Commutative	$x + y = y + x$	$xy = yx$
Distributive	$x(y + z) = xy + xz$	$x + yz = (x + y)(x + z)$
Complement	$x + x' = 1$	$xx' = 0$

Theorem	Statement
1	$x + x = x, \quad x \cdot x = x$
2	$x + 1 = 1, \quad x \cdot 0 = 0$
Involution	$(x')' = x$
Associative	$x + (y + z) = (x + y) + z, \quad x(yz) = (xy)z$
DeMorgan	$(x + y)' = x'y', \quad (xy)' = x' + y'$
Absorption	$x + xy = x, \quad x(x + y) = x$

(a) $F = A'BC + A'BC' + AB'C + ABC.$ [4 pts]

Answer:

$$\begin{aligned} F &= A'B(C + C') + AC(B' + B) && \text{(factor)} \\ &= A'B*1 + AC*1 && \text{(complement)} \\ &= A'B + AC && \text{(identity)} \end{aligned}$$

(b) $G = (x + y)(x + y')(x' + z).$ [4 pts]

Answer:

$$\begin{aligned} G &= [(x + y)(x + y')](x' + z) \\ &= [x + yy'] (x' + z) && \text{(distributive reverse)} \\ &= x(x' + z) = xx' + xz = 0 + xz = xz \end{aligned}$$

(c) $H = wx'y + wx'y' + wxy + w'xy.$ [4 pts]

Answer:

$$\begin{aligned} H &= wx'(y + y') + xy(w + w') && \text{(factor)} \\ &= wx'*1 + xy*1 && \text{(complement)} \\ &= wx' + xy && \text{(identity)} \end{aligned}$$

11. **Boolean identity proof.** Determine whether the following Boolean equation is true or false. Justify your answer using the postulates and theorems of Boolean algebra. Do not use a truth table. [8 pts]

$$AB + A'C + BC = AB + A'C.$$

```

Answer: TRUE.
AB + A'C + BC
= AB + A'C + BC(A + A')      (complement: A + A' = 1)
= AB + A'C + ABC + A'BC      (distributive)
= AB(1 + C) + A'C(1 + B)      (factor / associative)
= AB*1 + A'C*1                (x + 1 = 1)
= AB + A'C                    (identity)

```

12. **Logic gates and bitwise operations.** [10 pts]

- (a) Draw a gate-level logic diagram for

$$Y = (A + B')(C \oplus D) + A'C.$$

Use only NOT, AND, OR, and XOR gates. Label all inputs and the output Y . Do not simplify the expression before drawing. [6 pts]

- (b) Let $P = 10101100$ and $Q = 01110110$. Compute $P \oplus Q$, $P \text{ NAND } Q$, and $P \text{ NOR } Q$ bit by bit. [4 pts]

```

Answer:
P XOR Q  = 11011010
P AND Q  = 00100100, so P NAND Q = 11011011
P OR Q   = 11111110, so P NOR Q  = 00000001

```